

Durability of GFRP bars in the simulated marine environment and concrete environment under sustained compressive stress

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HIGHLIGHTS

- Axial compressive performance of GFRP bars under sustained stress was investigated.
- Stress level had an adverse effect on compressive strength retention.
- GFRP bars conditioned in salt and alkaline solution had different failure modes.
- A degradation model for GFRP bars under sustained compressive stress was proposed.

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ABSTRACT

The accelerated aging test results of the study on the compressive performance of glass fiber reinforced polymer (GFRP) bars under sustained compressive stress in the marine environment simulated by a salt solution and the concrete environment simulated by an alkaline solution were presented in this paper. The test parameters were sustained stress levels (0%, 20%, 40% of the ultimate compressive strength), exposure temperature (40 °C, 60 °C, 80 °C) and exposure time (16 d, 60 d, 90 d). The test results indicated that high temperature led to postcure of the resin matrix which was beneficial to the compressive performance of the GFRP bars. The conditioned GFRP bars failed due to the longitudinal splitting of the resin matrix or the fracture of the fibers and the debonding of the fiber/matrix interface. After immersing in the salt solution at 80 °C for 90 d, the strength retentions of the specimens with stress levels of 0%, 20% and 40% were 67%, 62% and 55% respectively. After immersing in the alkaline solution at 80 °C for 90 d, the strength retentions of the specimens with stress levels of 0%, 20% and 40% were 44%, 36% and 24% respectively. The compressive strength retention of the conditioned GFRP bars decreased with the increase of the sustained stress level, exposure temperature or exposure time. Increasing the exposure temperature or the sustained stress level could increase the degradation rate of the compressive strength of the GFRP bars. The compressive strength retention of the GFRP bars conditioned in the alkaline solution was lower than that in the salt solution. And the higher the temperature, the more obvious the detrimental effect of the stress level on the compressive strength of the GFRP bars was. Based on the test results, a degradation model for the compressive strength of the GFRP bars under sustained compressive stress in the simulated marine and the simulated concrete environments was proposed.

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1. Introduction

Corrosion of steel reinforcement caused the deterioration of the civil engineering projects and affected the service life of the civil infrastructures [1–3] when steel was used in aggressive environments (bridges, ports, chemical plants and water tanks). Fiber reinforced polymer (FRP) bars have emerged as the promising alternatives to traditional steel bars due to many advantages of

light weight, high strength and ultraviolet, especially their corrosion resistance [1,4,5].

Some researchers have investigated the durability of FRP bars in a marine environment or a concrete environment. The marine environment and the concrete environment were usually simulated by a salt solution [6] and an alkaline solution [7].

However, most of the researchers focused on the tensile performance of the FRP bars. Chen et al. [5,8] conducted experiments on the tensile properties of bare FRP bars and FRP bars embedded in the concrete exposed to different solutions (water, salt solution and alkaline solutions of different concentrations) at different

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temperature of 20 °C, 40 °C and 60 °C. After exposed to the alkaline solution at 40 °C and 60 °C, there were white blisters on the surface of the GFRP bars. The temperature of 40 °C did not have an obvious accelerating effect. The retention of the GFRP bars in simulated seawater was higher than that in the simulated pore solution of the concrete. Based on the test results of GFRP bars exposed to the alkaline solution at 20 °C, 40 °C and 60 °C, a prediction model for GFRP bars was proposed to evaluate the long-term durability performance in simulated concrete pore solution.

Some researchers investigated the tensile performance of the GFRP bars under sustained stress which was closer to field conditions. It was found that there were three stress corrosion mechanisms (stress dominated, crack propagation dominated, and diffusion dominated) of the GFRP bars under sustained load embedded in moist concrete. And the corrosion mechanisms were affected by the type of resin matrix, size, and sustained stress level of the FRP bars and pH of the alkaline environment [9]. The residual tensile strength of GFRP bars with the diameters of 16 mm, 12.7 mm and 9.5 mm under tensile stress levels equal to 19–38% of the guaranteed strength exposed to the alkaline solution was studied. After the GFRP bars exposed to the alkaline solution for 30 d–417 d at ambient temperature, the reductions in strength were 7–31.5% of the guaranteed strength [2,10]. The reductions in tensile strength of the GFRP bars under sustained stress levels equal to 0% and 25% of the ultimate tensile strength exposed to the alkaline solution for 183 d at 60 °C were 48.8% and 55.6% [11]. Wang et al. [12] investigated the durability of GFRP composite under sustained load exposed to the saltwater. The test results showed that the tensile properties of the GFRP bars degraded slightly after immersing for 180 d, but degraded significantly as the sustained load increased after immersing for 360 d. El-Hassan et al. [13] investigated the durability of two types of GFRP bars embedded in moist seawater-contaminated concrete exposed to tap water. The strength retentions of the two types of GFRP bars under sustained stress of 25% of the ultimate tensile strength were 84% and 70% at 20 °C, 15% and 12% at 60 °C. Fergani et al. [14] investigated the tensile property of the GFRP bars subjected to different levels of sustained load in moist concrete, an alkaline solution and tap water. The test results indicated that the higher temperature and higher level of sustained stress accelerated the degradation of the strength. The strength degraded due to the high diffusion of the solution to the resin matrix.

With the improvement of the FRP materials, they were expected to be used as the compressive reinforcements [1,15–17]. However, there was very little research on the compressive performance of the FRP bars, especially the durability of the FRP bars under sustained compressive stress in the marine environment or the concrete environment. Fu et al. [18] tested 90 GFRP bars after immersing in an alkaline solution for 183 d. The reductions in compressive strength were 29.6%, 39.1% and 47.6% at 40 °C, 60 °C and 80 °C. The SiO₂ content of the GFRP bars reduced from 62.1% to 52.1%, 50.7% and 47.7% at 40 °C, 60 °C and 80 °C respectively. Sun et al. [19] reported that the residual compressive strength of the GFRP bars with the diameters of 8 mm, 10 mm and 12 mm were 65.0%, 67.1% and 76.9% of the ultimate strength after immersing in a salt solution for 150 d. Deng et al. [20] investigated the compressive performance of GFRP bars combined the effect of solution type, exposure temperature and stress level. The test results demonstrated that the sustained stress and temperature accelerated the degradation of the compressive strength. The reductions in compressive strength of the GFRP bars under sustained stress of 0% and 40% of the ultimate compressive strength were 31.8% and 44.2% in the alkaline solution at 60 °C, 22.2% and 29% in the salt solution at 60 °C.

GFRP bars were widely used in civil engineering for their high cost-effective performance. So GFRP bars were selected to be inves-

tigated in this study. This paper aimed to provide a better understanding of the degradation of the GFRP bars under sustained compressive stress in the marine environment and concrete environment and provide a degradation model to predict the long-term compressive performance of the GFRP bars.

2. Experimental program

2.1. Material and specimens

Fig. 1 showed the GFRP bars used in this experiment. The diameter of the GFRP bars was 10 mm. The GFRP bars were made of continuous E-glass fibers impregnated in epoxy resin using the pultrusion process. The fiber volume fraction was 70%. During the pultrusion of the GFRP bars, the surface was ribbed and helically wrapped with a glass fiber bundle. The ultimate compressive strength of the GFRP bars was 420.4 MPa tested according to [21].

A total of 165 specimens were prepared in this experiment. The length-diameter ratio of the compressive specimens was 2.5 according to [21]. As shown in Fig. 2, each specimen consisted of two 20 mm long steel tubes with an outside diameter of 27 mm and a GFRP bar with an unbraced length of 25 mm. The steel tubes were designed as collars at the end regions of the specimens to prevent premature failure of the GFRP bars. The gaps between the GFRP bar and the tubes were grouted with epoxy resin matrix.

2.2. Pre-stress procedure

The procedure of specimens subjected to sustained compressive stress was presented as follows:

Firstly, three GFRP specimens with a 0.5 mm thickness rubber shim on each top were placed in the grooves of the top and bottom stainless steel plates as shown in Fig. 3(c). Inserting the rubber shims was to relieve the uneven stress distribution because of the slight differences in length of the three specimens. Secondly, the pre-stress setup and the GFRP specimens were simultaneously



Fig. 1. Unconditioned GFRP bar.

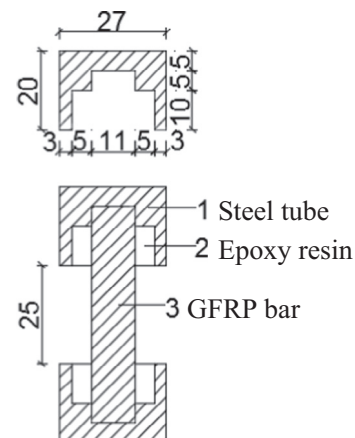


Fig. 2. The compressive specimen of GFRP bar.

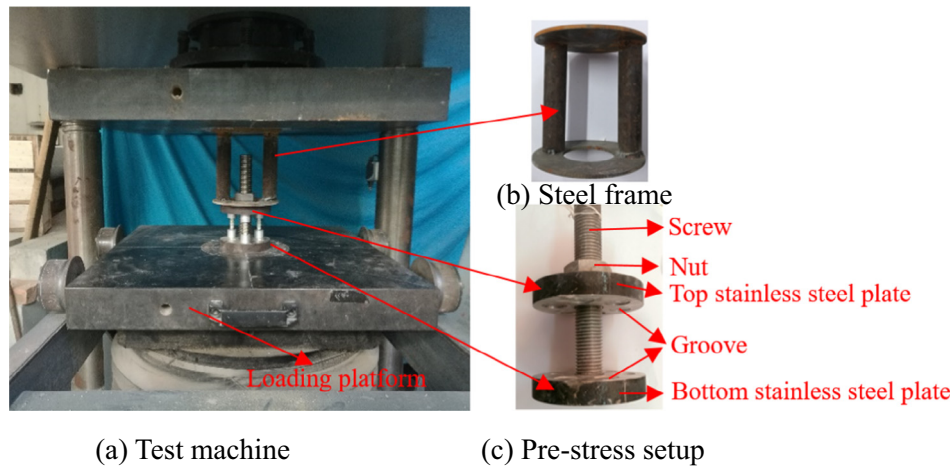


Fig. 3. Imposing the sustained stress on the GFRP specimens.

placed on the platform of the testing machine as shown in Fig. 3(a). The steel frame in Fig. 3(b) was placed on the top stainless steel plate. Thirdly, the axial load on the pre-stress setup was applied by the electrohydraulic servo pressure testing machine. When the applied load reached the required value which measured by the data logger, the location of the loading platform was kept unchanged. Then tightened the nut on the screw using a wrench. At last, the specimens were under sustained compressive stress. The test machine was unloaded. The GFRP specimens and the pre-stress setup were removed from the platform and exposed to the solutions simulated to the marine environment and the concrete environment.

2.3. Simulated marine environment and simulated concrete environment

The marine environment was simulated by a salt solution composed of 24.53 g of NaCl, 4.09 g of Na_2SO_4 , 1.16 g of CaCl_2 and 0.69 g of KCl in 1 L of distilled water according to [6]. The concrete environment was simulated by an alkaline solution composed of 118.5 g of $\text{Ca}(\text{OH})_2$, 0.9 g of NaOH, and 4.2 g of KOH in 1 L of deionized water according to [7]. The alkaline solution had a pH value of 12.6–13. The solution temperature was designed to be 40 °C, 60 °C and 80 °C for the accelerated aging test. The specimens under sustained compressive stress equal to 0%, 20% and 40% of the ultimate compressive strength were exposed to the prepared solutions in water tanks for 16 d, 60 d and 90 d.

2.4. Test procedure

After reaching the required immersing time in the solutions, the specimens were removed from the pre-stress setup. Each specimen was subjected to the compressive test using the YAW6306 electrohydraulic servo pressure test machine produced by American MTS company. The loading duration time of each specimen was 2–4 min.

3. Experimental results and discussion

FRP bars were heterogeneous of fibers and resin matrix. The resin acted as a matrix bonding the fibers together and spreading the load applied to the FRP bars between the individual fibers. The resin protected the fibers from corrosion in the aggressive environment. The deterioration of the performance of the fibers,

resin or their interface would weaken the entire performance of the FRP bars.

Water molecule, chloride ion (Cl^-) and hydroxyl ion (OH^-) in the solutions induced the degradation of the GFRP bars. The degradation process involved physical and chemical changes.

The sustained stress level, exposure temperature and exposure time were three factors that influenced the degree and rate of the degradation of the compressive performance of the GFRP bars.

3.1. Scanning electron microscopy (SEM)

SEM observation and image analysis were performed on some selected specimens using an FEI Quanta 200 SEM. The SEM images of the cross-section and longitudinal section of the samples were presented to investigate the degradation mechanism.

Fig. 4 showed the SEM images of the unconditioned GFRP bars. It was found that the fibers, resin and the fiber-resin interface were all intact.

Fig. 5(a) showed the SEM images of the specimen 90-80 °C-0% in the salt solution. At the edge of the specimen, there was only a little damaged area. But there were two microcracks observed in the cross-section. The degradation of the specimen was induced by the penetration of the water molecules and Cl^- . Water molecule was a low molecular weight material, which could space the polymer chains once the water molecular infused into the matrix [22]. Cl^- in the salt solution diffused inside the resin inducing the internal stress. With the increasing internal stress, micro cracks appeared in the FRP bars and the fiber-resin cohesion degraded. Cl^- penetrated into the GFRP bars through the cracks accelerating the debonding between the fibers and the resin.

Fig. 5(b) showed the SEM images of the specimen 90-80 °C-40% in the salt solution. Compared Fig. 5(b) with Fig. 5(a), it was clear

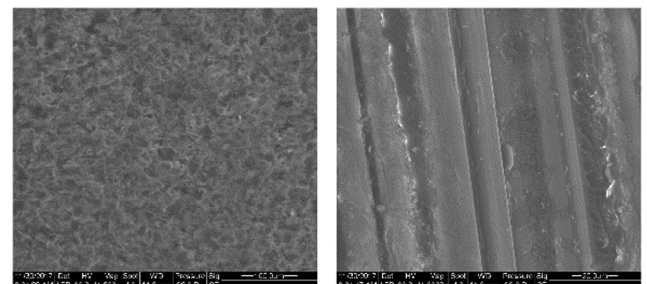


Fig. 4. Unconditioned specimen.

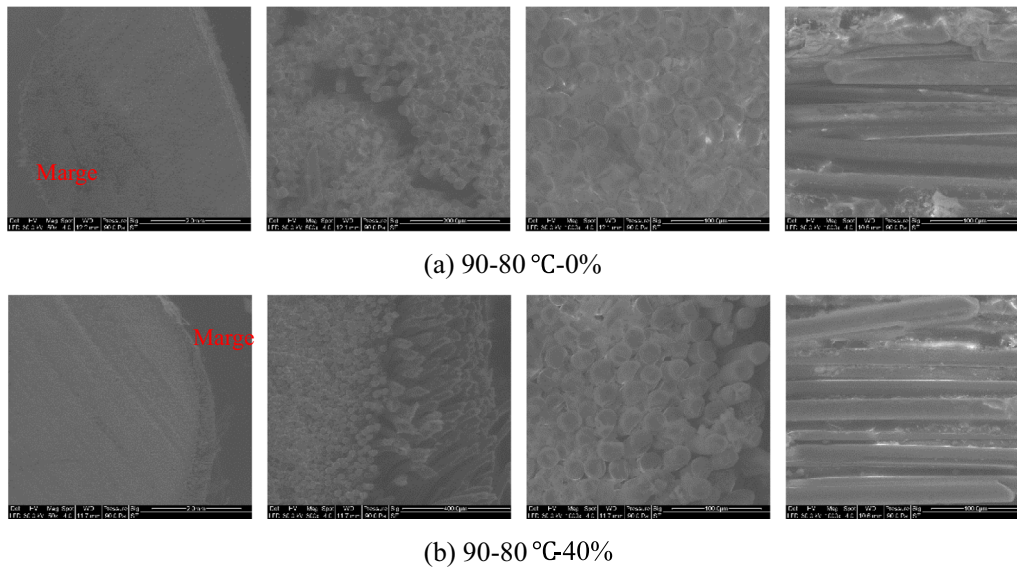


Fig. 5. Specimens conditioned in the salt solution.

that the damaged area of specimen 90-80 °C-40% was larger than that of specimen 90-80 °C-0%. The higher stress that the specimen 90-80 °C-40% sustained caused the higher debonding stress across fiber-resin interface and accelerated the diffusion of the solution. As the hydrolysis degree of the resin increased, the voids began to form. The surface of the fiber in the damaged area were smooth and the resin around the fibers was less. There was an obvious boundary between the damaged and undamaged area of the specimen 90-80 °C-40% and the fibers in the damaged area were looser.

Fig. 6(a), (b) and (c) showed the SEM images of the specimens 16-80 °C-40%, 90-80 °C-0% and 90-80 °C-40% in the alkaline solution respectively. The degradation of these specimens was induced by the penetration of the water molecules and OH^- . When epoxy was mixed with anhydride curing agent, ester groups appeared in the cured resin. OH^- in the alkaline solution affected the performance of the GFRP bars by reacting with the ester group in the resin as presented in Eq. (1). The ester linkages were broken inducing the degradation of the resin and the debonding of fiber/matrix interface.

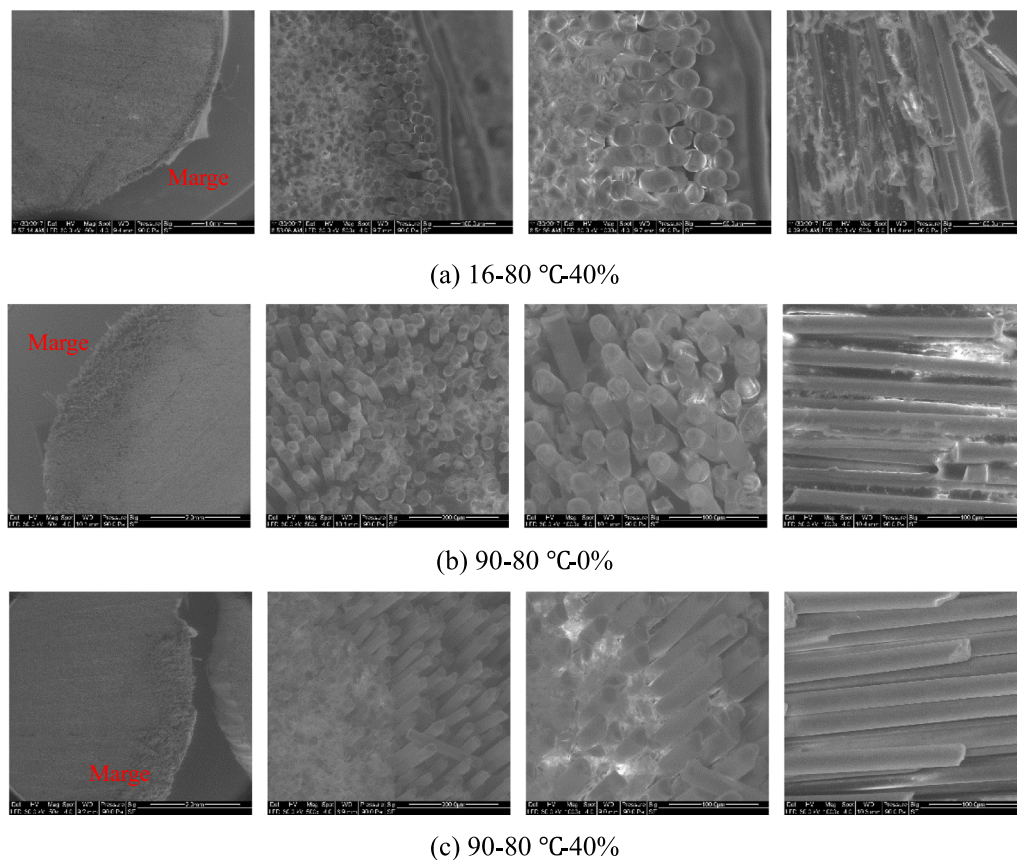
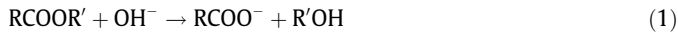


Fig. 6. Specimens conditioned in the alkaline solution.



Compared Fig. 6(c) with Fig. 6(a), the damaged area significantly extended from the edge to the core of the GFRP bars in the radial direction when the exposure time increased from 16 d to 90 d. It demonstrated that as the exposure time increased, the penetration depth of the solution increased and the hydrolysis degree of the resin increased. There were obvious boundaries between the damaged and undamaged area in both the specimens 16-80 °C-40% and 90-80 °C-40%. Compared with the fibers in the specimen 90-80 °C-40%, the fibers in the specimen 16-80 °C-40% had more resin coverage.

Compared Fig. 6(c) with Fig. 6(b), the damaged area size of the specimens 90-80 °C-40% and 90-80 °C-0% was similar. In the specimen 90-80 °C-0%, there was a transition region between the damaged area and undamaged area. However, in the specimen 90-80 °C-40%, there was a distinct boundary between the damaged area and undamaged area. Comparing with the fibers in specimen 90-80 °C-0%, the fibers in specimen 90-80 °C-40% had less resin coverage. The fibers in the specimen 90-80 °C-40% were looser than that in the specimen 90-80 °C-0%.

Compared Fig. 6(b) with Fig. 5(a), Fig. 6(c) with Fig. 5(b), it was concluded that the GFRP bars in the alkaline solution had a faster degradation than that in the salt solution.

According to the above analysis, among the specimens conditioned in the solutions at 80 °C for 90 d, only the specimens with the stress level equal to 40% had obvious boundaries between the damaged area and undamaged area. It demonstrated that the high stress level accelerated the diffusion of the solutions into the GFRP bars which could promote the hydrolysis of the resin.

The SEM images above showed that no corrosion pits were observed on the fiber surface after conditioning in the salt or the alkaline solution in this experiment. The fibers were protected well by the resin.

3.2. Surface appearance

The specimen code was made up of numbers represented the exposure time, exposure temperature and stress level in sequence. For example, 16-40 °C-20% represented that the GFRP specimen under sustained compressive stress equal to 20% of the ultimate compressive strength was exposed to the solution at 40 °C for 16 d.

Fig. 1 showed that the unconditioned GFRP bar had a smooth and lustrous surface appearance.

Figs. 7 and 8 showed the surface appearance of the GFRP bars exposed to the salt solution and the alkaline solution respectively. The stress level slightly affected the appearance of the GFRP bars when the exposure temperature and exposure time kept unchanged whether in the salt solution or the alkaline solution. As the exposure temperature or exposure time increased, the smoothness and the luster of the conditioned GFRP bars exposed to the salt solution and the alkaline solution decreased. The surface appearance color of the GFRP bars exposed to the salt solution changed slightly, while that of the GFRP bars exposed to the alkaline solution became more whiter. There were no obvious cracks or pits on the surface of the conditioned GFRP specimens in both salt solution and alkaline solution.

3.3. Compressive failure modes of the GFRP bars

Figs. 9 and 10 showed the failure modes of the unconditioned specimens and the specimens conditioned in the salt solution. It could be seen that the failure modes of the unconditioned specimens and specimens conditioned in the salt solution were similar. For these specimens, when the load was close to the ultimate compressive capacity, some short cracks appeared between the two adjacent ribs (such as AB and CD in Fig. 10(c)). The short cracks between the adjacent ribs were connected by the cracks along

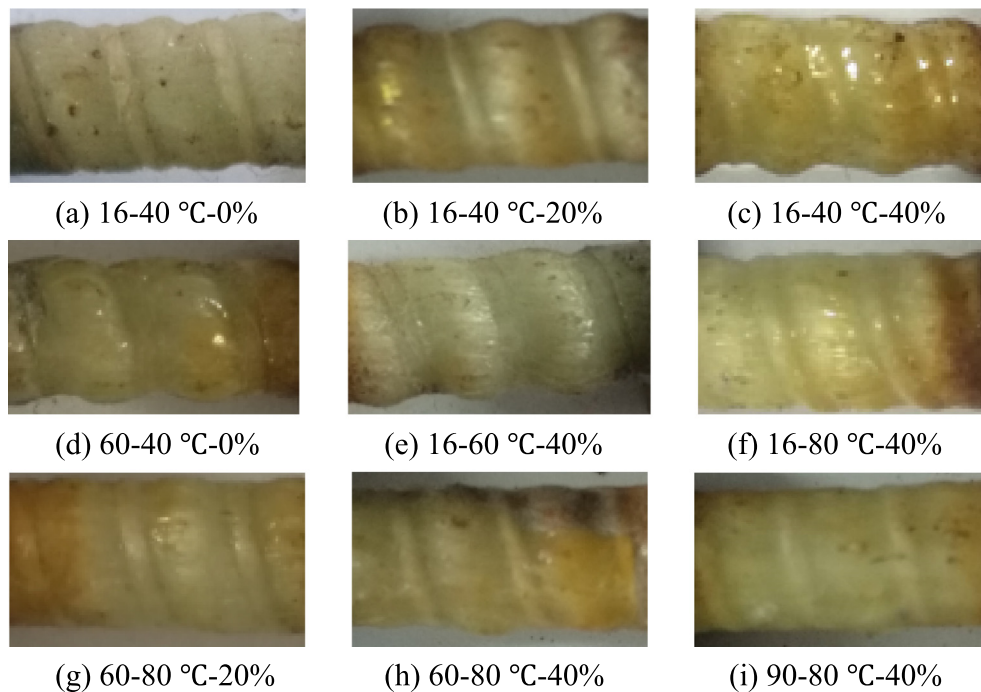


Fig. 7. Surface appearance after immersing in the salt solution.

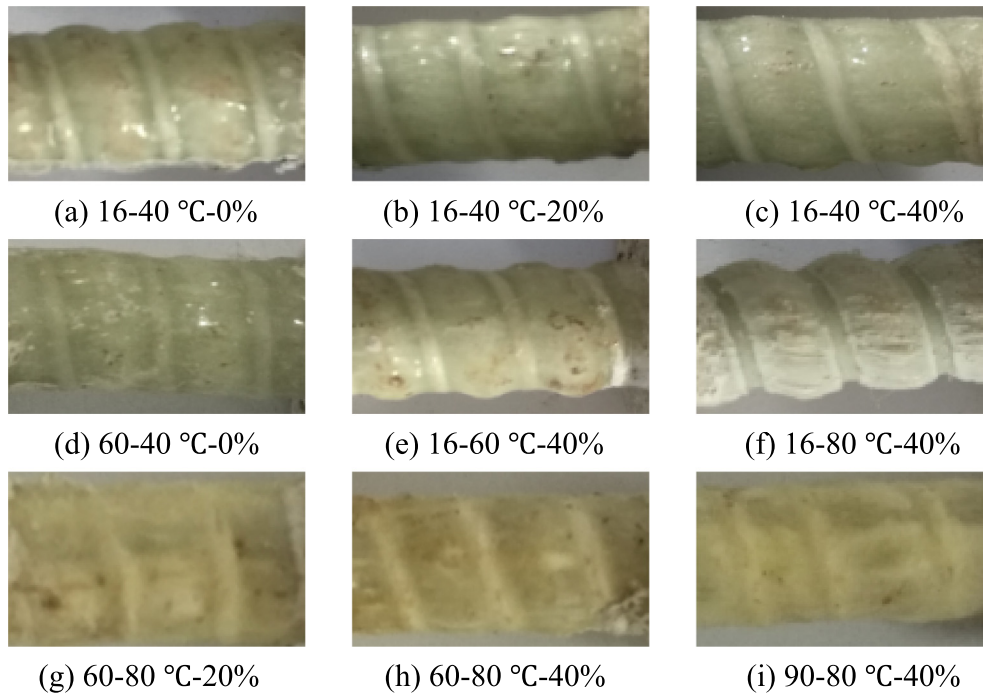


Fig. 8. Surface appearance after immersing in the alkaline solution.



Fig. 9. Failure mode of unconditioned specimen.

the rib (such as BC in Fig. 10(c)) resulting diagonal cracks on the whole. The failure of the unconditioned specimens and specimens conditioned in the salt solution occurred due to the longitudinal splitting of the resin matrix. There was no obvious fibers loosening in the failed specimens immersing in the salt solution. It demonstrated that the resin was not seriously corroded by the water molecule and Cl^- , and the bond strength between fiber and resin of the tested specimens did not degraded obviously.

Fig. 11 showed the failure modes of the specimens conditioned in the alkaline solution. As shown in Fig. 11(a), the specimens exposed to the alkaline solution for 16 d had a similar failure mode to that of the unconditioned specimen. It demonstrated that when

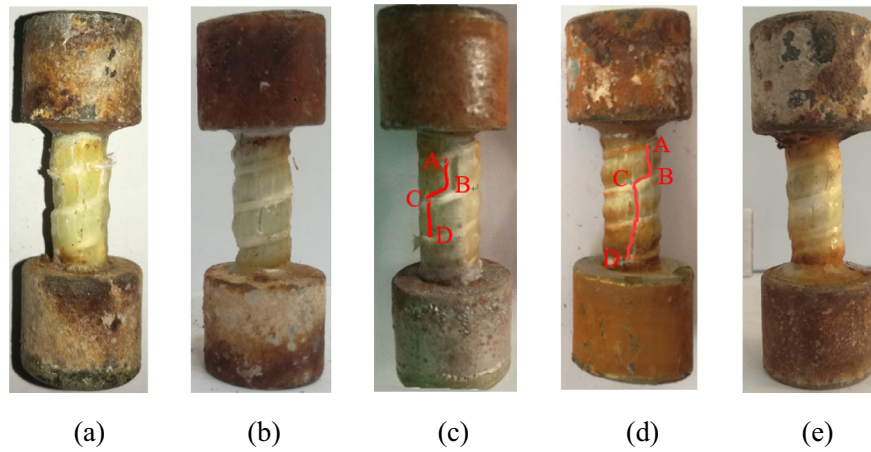
the exposure time was less than 16 d, the water molecule and OH^- had less corrosion effect on the GFRP bars.

The other specimens conditioned in the alkaline solution for 60 d and 90 d had a significantly different failure mode shown in Fig. 11(b)–(e). The failure of these specimens occurred in the form of a brush-like appearance. The resin of these specimens had a great degradation due to the hydrolysis induced by the water molecular and OH^- . The debonding between the fibers and the resin increased. The failure was accompanied by the delamination of the fibers and the resin. The fibers of the failed specimens undergone an obvious deformation. As the exposure temperature or exposure time increased, the delamination degree of the fibers and the resin became more obvious. The failure of these specimens was due to the fracture of the fibers and the degradation of the fiber/matrix interface.

4. Effect of test parameters on the compressive strength of the GFRP bars

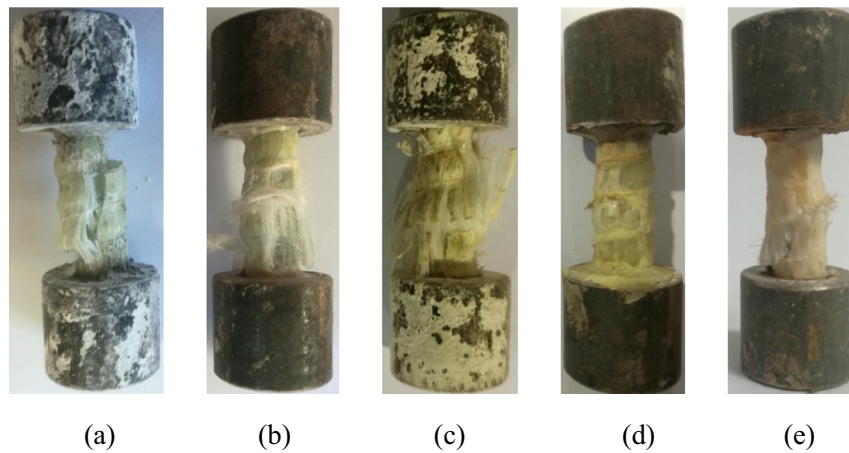
Fig. 12(a), (b) and Fig. 13(a), (b) showed the compressive strength and the compressive strength retentions of the GFRP bars in the salt solution and the alkaline solution. In the legends in Figs. 12 and 13, the first number and the second number represented the exposure temperature of the GFRP bars and the stress level that the GFRP bars sustained. For example, the legend 40 °C-20% represented the specimen conditioned in the solution at 40 °C and sustained stress equal to 20% of the ultimate compressive strength of the GFRP bars.

As shown in Fig. 12(a), after immersing in the salt solution for 16 d, the compressive strength of the specimens 40 °C-20%, 60 °C-0%, 60 °C-20%, 80 °C-0% and 80 °C-20% were 2%, 8%, 3%, 15% and 12% higher than that of the unconditioned specimen. As shown in Fig. 13(a), after immersing in the alkaline solution for 16 d, the compressive strength of the specimens 80 °C-0% and 80 °C-20% were 6.1% and 3% higher than that of the unconditioned specimen. The temperature had dual effects on the mechanical properties of the GFRP bars. One effect was the postcure of the resin attributed



(a) 16-80 °C-40% (b) 60-60 °C-20% (c) 60-80 °C-40% (d) 90-60 °C-20% (e) 90-80 °C-40%

Fig. 10. Failure modes of specimens conditioned in the salt solution.



(a) 16-80 °C-40% (b) 60-60 °C-20% (c) 60-80 °C-40% (d) 90-60 °C-20% (e) 90-80 °C-40%

Fig. 11. Failure modes of specimens conditioned in the alkaline solution.

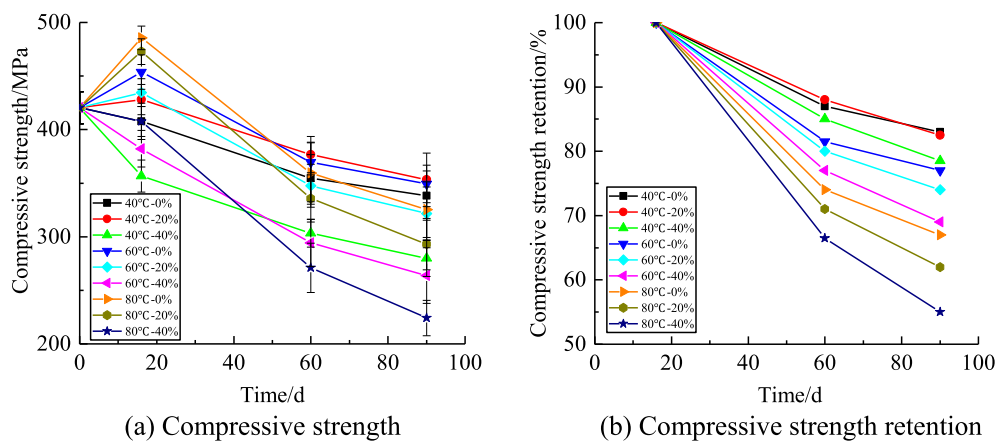


Fig. 12. Test results of the GFRP bars conditioned in the salt solution.

to the elevated temperature. The postcure process occurred in the initial stage of the immersing time and was beneficial to the mechanical performance of the GFRP bars [12,23]. The higher the temperature was, the more sufficient the postcure process was.

The other effect was the accelerated corrosion of the solution due to the elevated temperature. The compressive strength was determined by the net effect of the temperature on the mechanical performance.

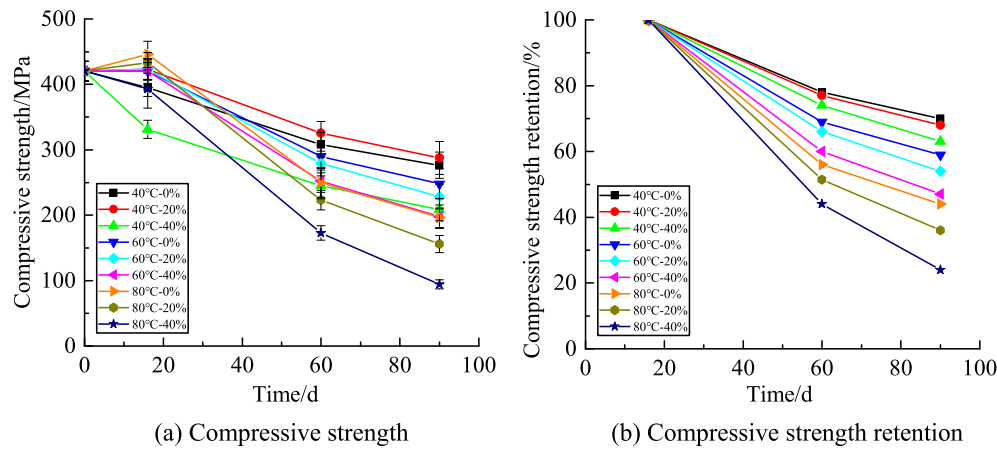


Fig. 13. Test results of the GFRP bars conditioned in the alkaline solution.

After immersing in the salt solution or the alkaline solution for 16 d, the compressive strength of all the specimens degraded with the increasing exposure time. To eliminate the different influences of the postcure effect on the different specimens, the compressive strength of the specimen immersing for 16 d was taken as a reference value. The strength retention of each specimen exposed for more than 16 d was defined as the ratio of degraded strength to its reference value (compressive strength at 16 d) in this paper. Thus the compressive strength retention of each specimen aging for 16 d was 100%.

As shown in Fig. 12(b) and Fig. 13(b), after 90 d of exposure, the compressive strength retentions of all the conditioned specimens decreased with the increasing exposure time. The compressive strength retentions of the specimens conditioned in the salt solution lost 17–45%, and that conditioned in the alkaline solution lost 30–76%. This was because the solution permeated into the specimens which induced the hydrolysis of the resin and the debonding of the interface between the fiber and the resin over time. It was also observed that the degradation rate after 60 d of exposure was lower than that between 16 d and 60 d.

When the sustained stress level increased from 0% to 40%, the amplitude of degradation of the specimens immersing for 90 d in the salt solution were 4.5% at 40 °C, 8% at 60 °C and 12% at 80 °C, and that in the alkaline solution were 7% at 40 °C, 12% at 60 °C and 20% at 80 °C. It indicated that the compressive strength retentions decreased with the increasing sustained stress level. The reason was that the sustained stress accelerated the propagation of the micro cracks in the specimens. It was also concluded that the detrimental effect of the sustained stress level on the compressive strength became apparent at elevated temperature.

Compared the compressive strength retentions of the specimens immersing for 90 d at 80 °C with that at 40 °C, for the specimens under the stress level of 0%, 20% and 40%, the reductions of the compressive strength retentions were 16%, 20.5% and 23.5% in the salt solution, and were 26%, 32% and 39% in the alkaline solution. The higher the exposure temperature, the lower the compressive strength retention. The result indicated that the higher temperature made the solution permeate into the specimens more easily. And the adverse effect of the temperature on the strength retention was more obvious at higher stress level than that at lower stress level.

5. Prediction of the compressive strength degradation

The existing prediction models for long-term performance of FRP bars mainly based on Fick's law [24–26] or Arrhenius theory

[5,27,28]. However, the Fickian diffusion law was only applicable to the prediction of the short-term data. Because with the increasing immersing time, the cracks generated and propagated through the polymer matrix and the fiber-resin interface debonded due to the solution diffusion and the sustained stress. The way in which the solution entered into FRP bars changed from entire diffusion (due to concentration gradient) to flow along cracks, so Fick's law was not applicable [29]. The basic assumption of Arrhenius theory was that the degradation mechanism of the material did not change with exposure temperature and time [30]. If temperature was not the only factor that controlled the degradation mechanism of the GFRP composite, Arrhenius theory was not justifiable [31].

In this experiment, the combined temperature and sustained compressive stress affected the degradation process of the GFRP bars. The Fick's law or Arrhenius theory could not be directly used to predict the long-term performance of the GFRP bars. And so far, there were no prediction models considering the coupled effects of sustained compressive stress and temperature.

Based on the previous prediction models for the long-term performance [3,12,25,30,32,33] of the FRP composites and the current experimental data, a new prediction model for the long-term compressive strength of GFRP bars considering sustained compressive stress and temperature was proposed as follows:

$$Y = A \log t - 1.2A + 100 \quad (2)$$

where Y was the compressive strength retention, %; t was immersing time, d; A was the parameter for calculating compressive strength retention related to sustained compressive stress level and temperature.

5.1. Prediction model for the GFRP bars in the salt solution

In Fig. 14, the scatters were the experimental data and the curves were the prediction models.

The parameter A in Eq. (2) for the GFRP bars in the salt solution was calculated as follows:

$$A = -0.469T - 0.075T \times \exp(3\sigma) \quad (3)$$

where T was the exposure temperature, °C; σ was the stress level that the GFRP bars sustained in the salt solution.

Fig. 15 showed the experimental and predictive compressive strength retentions of the GFRP bars immersing in the salt solution for 60 d and 90 d. The ratios of the experimental value to the predictive value were 0.98–1.03. The proposed model agreed well with the experimental results.

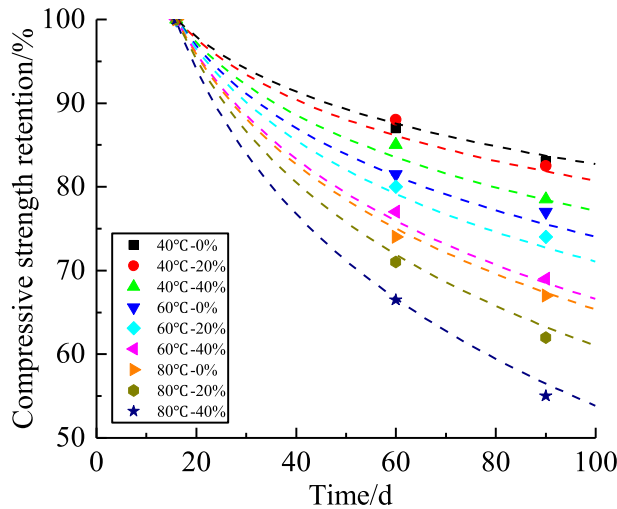


Fig. 14. Experimental compressive strength retentions and prediction models for the GFRP bars in the salt solution.

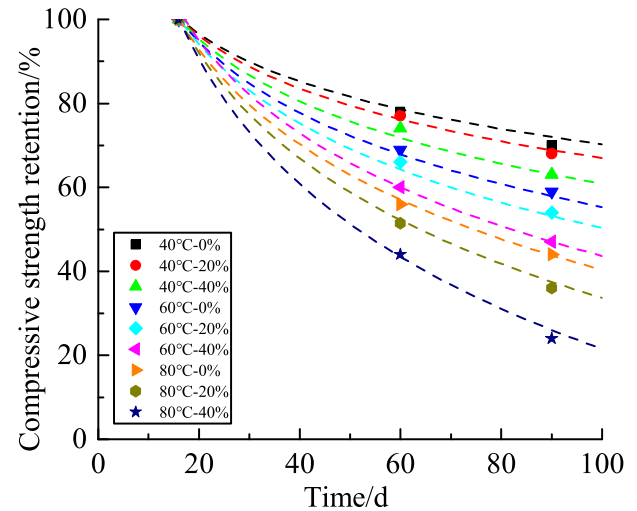


Fig. 16. Experimental compressive strength retentions and prediction models for the GFRP bars in the alkaline solution.

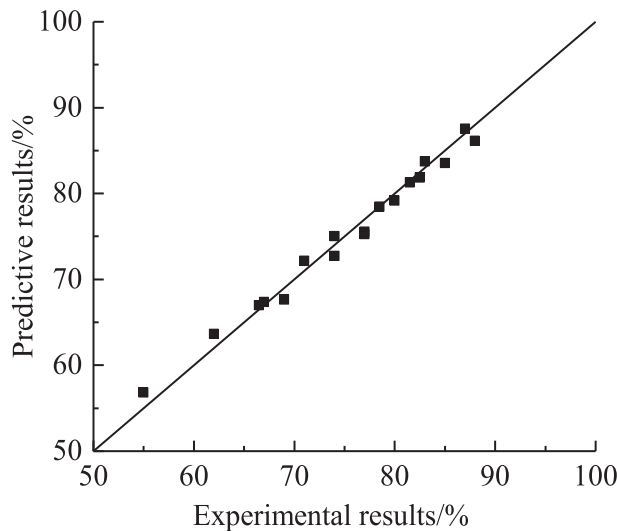


Fig. 15. Experimental and predictive compressive strength retentions of the GFRP bars in the salt solution.

The experimental results in the literature [19] were used to verify the rationality of the proposed model. The experimental strength retentions were 86% and 79% of the GFRP bars immersing in the salt solution for 90 d and 120 d. The predictive values calculated by Eq. (2) were 92% and 90%.

In 2017, the average annual temperature in Beijing was 12.6 °C. According to the proposed model, the compressive strength retentions of the GFRP bars under sustained compressive stress levels equal to 20% and 50% were 77% and 69% after serving for 50 years in the marine environment.

5.2. Prediction model for the GFRP bars in the alkaline solution

In Fig. 16, the scatters were the experimental data and the curves were the prediction models.

The parameter A in Eq. (2) for the GFRP bars in the alkaline solution was calculated as follows:

$$A = -0.817T - 0.123T \times \exp(3\sigma) \quad (4)$$

Fig. 17 showed the experimental and predictive compressive strength retentions of the GFRP bars immersing in the alkaline

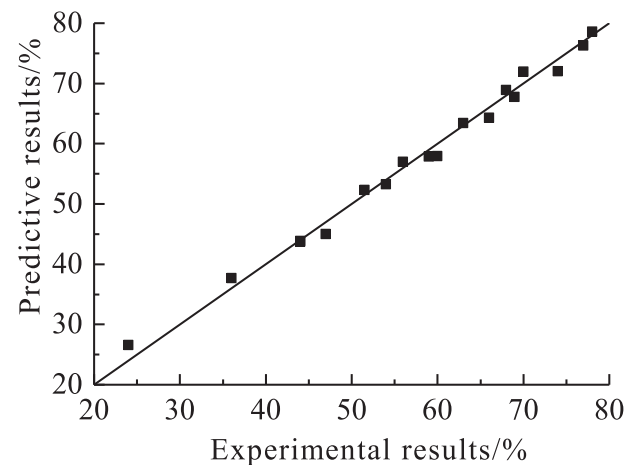


Fig. 17. Experimental and predictive compressive strength retentions of the GFRP bars in the alkaline solution.

solution for 60 d and 90 d. The ratios of the experimental value to the predictive value were 0.97–1.11. The proposed model agreed well with the experimental results.

The experimental results in the literature [18,20] were used to verify the rationality of the proposed model. The comparison results were shown in Fig. 18.

In 2017, the average annual temperature in Beijing was 12.6 °C. According to the proposed model, the compressive strength retentions of the GFRP bars under sustained compressive stress levels equal to 20% and 50% were 59% and 47% after serving for 50 years in the alkaline environment. Eqs. (3) and (4) showed that the GFRP bars in the alkaline solution had a faster degradation compared with the GFRP bars in the salt solution.

The GFRP bars used in field concrete structures were wrapped in concrete which was different to a certain extent from the bare GFRP bars immersing in the alkaline solution [34]. When the GFRP bars embedded in concrete deformed due to compression or moisture absorption, the concrete could provide constraint on the GFRP bars [35]. The propagation rate of the cracks slowed down and the diffusion rate of the solution became slower, which were beneficial to the compressive strength. Therefore the proposed model for GFRP bars conditioned in the alkaline solution was conservative in predicting long-term strength.

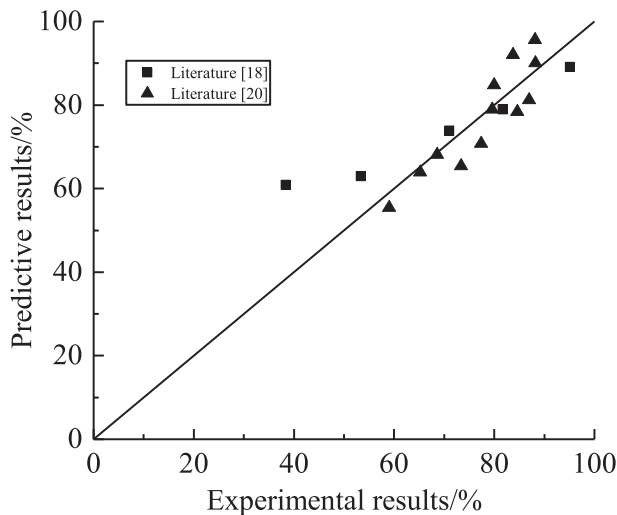


Fig. 18. The experimental compressive strength retentions in the literature [18,20] and the predictive results.

6. Conclusions

- 1) With the increasing stress level, the damaged area of the GFRP bars immersing in the salt solution increased, while the damaged area size changed slightly in the alkaline solution. But with the increasing stress level, the resin in the damaged area of the GFRP bars immersing in the alkaline solution had a more sufficient hydrolysis reaction. The damaged area of the GFRP bars immersing in the alkaline solution was larger than that immersing in the salt solution when the specimens had the same stress level, exposure temperature and exposure time.
- 2) Stress levels had little effect on the surface appearance of the GFRP bars. But with the increasing exposure temperature or the exposure time, the smoothness and the luster of the conditioned GFRP bars exposed to the salt solution and the alkaline solution decreased obviously.
- 3) The unconditioned GFRP bars and the GFRP bars exposed to the salt solution failed due to the longitudinal splitting of the resin matrix. The GFRP bars immersing in the alkaline solution for 16 d failed for the same reason as that of the unconditioned GFRP bars. The specimens immersing in the alkaline solution for 60 d or 90 d failed in the form of a brush-like appearance due to the fracture of the fibers and the debonding of the fiber/matrix interface.
- 4) After immersing in the salt solution at 80 °C for 90 d, the strength retentions of the specimens with the stress levels of 20% and 40% were 7.5% and 17.3% lower than that with the stress level of 0%. After immersing in the alkaline solution at 80 °C for 90 d, the strength retentions of the specimens with the stress levels of 20% and 40% were 18.2% and 45.5% lower than that with the stress level of 0%.
- 5) As the sustained stress level, exposure temperature or exposure time of the conditioned GFRP bars increased, the compressive strength retentions of the conditioned GFRP bars decreased. The sustained stress and the temperature had an accelerated effect on each other in terms of the degradation process of the GFRP bars.
- 6) A degradation model for GFRP bars considering sustained stress level and exposure temperature exposed to the salt solution and the alkaline solution was proposed. The model agreed well with the experimental results and could predict the long-term compressive strength retentions of the GFRP bars in the salt solution and the alkaline solution.

Declaration of Competing Interest

None.

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